

A CLINICIAN-VALIDATED LLM PIPELINE FOR STRUCTURING ONCOLOGY DATA AND CLINICAL TRIAL ELIGIBILITY MATCHING

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Abstract—Oncology departments generate large volumes of unstructured clinical data, complicating patient–trial matching and contributing to the under-enrolment of eligible patients in clinical trials. While Large Language Models (LLMs) offer promising solutions, existing approaches often rely exclusively on probabilistic reasoning, limiting transparency and reproducibility. This work proposes an integrated, hybrid system that leverages LLMs to transform unstructured oncology notes into structured, clinician-validated profiles and support transparent eligibility assessment.

The system operates in two core phases: (i) an LLM-based pipeline that extracts and normalizes clinical parameters from free-text notes for expert validation, and (ii) an automated module that converts trial criteria into machine-readable logical rules. Eligibility is then evaluated through a hybrid mechanism, combining deterministic rule-based execution for structured fields with LLM-assisted reasoning for complex, unstructured clinical context. This architecture balances the auditability of rule-based logic with the flexibility of generative models.

The system was evaluated using three open-source 8B-parameter LLMs across five pipeline tasks. Structured extraction yielded strong results, with the best model achieving 92.3% accuracy in criteria extraction and 78.3% in patient profile extraction. Transitioning clinical language into formal logic proved the most challenging task, with strict accuracies between 52.2% and 64.1%. Nevertheless, the final patient–trial matching stage demonstrated clinically reliable performance; the top models achieved up to 84.0% accuracy and expert-rated justification quality scores of 4.93 out of 5, ensuring decisions were grounded in clinical evidence. These findings demonstrate that modular pipelines combining structured representations, deterministic logic, and lightweight LLMs can support transparent, efficient, and clinically aligned patient recruitment in real-world oncology settings.

Index Terms—Large Language Models; Information Extraction; Clinical Trial Eligibility; Un structured Clinical Data; Clinician-in-the-Loop; Deterministic Eligibility Evaluation; Survival Analysis; Structured Patient Data

INTRODUCTION

Oncology departments generate large volumes of unstructured clinical data through free-text notes, making it difficult to efficiently retrieve clinically relevant information for decision support and research workflows. This challenge is particularly

evident in the context of clinical trial enrollment, where clinicians must manually extract, interpret, and compare patient information against extensive eligibility criteria. The process is time-consuming, cognitively demanding, and frequently interrupted by routine clinical activities, contributing to the persistent under-enrollment of eligible patients in clinical trials and limiting access to innovative therapeutic options.

Recent advances in Large Language Models (LLMs) have shown promise for clinical information extraction and patient–trial matching. However, existing approaches typically address these tasks independently, often relying on partially structured data or probabilistic LLM reasoning to assess eligibility. Such methods may perform well in controlled settings but lack the transparency, reproducibility, and auditability required in real-world healthcare environments. Moreover, limited attention has been given to integrated workflows that combine extraction from highly unstructured oncology diaries, clinician validation, logical formalization of eligibility criteria, and transparent eligibility assessment mechanisms that combine structured evaluation with LLM-assisted reasoning within operational hospital systems.

To address these limitations, this work proposes an integrated clinical decision-support system that leverages LLMs to transform unstructured oncology notes into structured, clinician-validated patient profiles and to enable transparent and reproducible eligibility assessment. The system includes an LLM-based extraction pipeline for identifying and normalizing clinically relevant parameters, a semi-automated process for extracting and formalizing trial inclusion and exclusion criteria into machine-readable logical rules, and a hybrid matching mechanism that evaluates validated patient profiles against validated eligibility criteria through a combination of deterministic rule-based assessment and LLM-assisted reasoning for criteria that cannot be directly represented within the structured patient schema. Clinician-in-the-loop validation is incorporated throughout the workflow to ensure correctness, interpretability, and trustworthiness.

This design allows structured and objectively measurable

criteria to be evaluated transparently through logical rules while preserving the ability to assess complex or schema-independent eligibility requirements using clinical context extracted from patient records.

The experimental evaluation, conducted using three lightweight open-source LLMs (8B parameters), demonstrated the feasibility of the proposed approach across all pipeline components. Results indicate a consistent performance pattern: models performed strongest in structured extraction tasks, faced greater difficulty in the logical formalization of eligibility criteria, and recovered competitive performance in the final eligibility assessment stage when grounded in explicit clinical context. Notably, expert clinical evaluation of the generated outputs confirmed that model decisions were consistently grounded in available evidence, with justification quality scores above 4.8 out of 5 for the best-performing models, supporting the clinical reliability of the proposed framework.

This article presents the system architecture, describes the extraction and matching pipelines, and reports an experimental evaluation of multiple LLMs in a real hospital context. The proposed approach aims to reduce administrative burden, improve traceability, and enhance the efficiency and reproducibility of clinical trial enrollment in oncology settings.

RELATED WORK

Research on clinical information extraction using Large Language Models (LLMs) has largely focused on transforming unstructured clinical narratives into structured representations through tasks such as Named Entity Recognition, [11], and attribute extraction, [9] and [1]. Existing approaches commonly rely on prompt-engineering strategies, including zero-shot, and few-shot prompting, as well as parameter-efficient fine-tuning techniques such as PEFT and Retrieval-Augmented Generation, [12], [10], [2], [11] and [8]. Hybrid frameworks that combine multiple techniques have also emerged, demonstrating improved robustness and extraction accuracy when evaluated against expert-annotated gold standards, [3] and [9]. Overall, the literature shows that LLMs can outperform traditional NLP pipelines and rule-based systems, offering greater flexibility and reduced development effort, [6] and [12]. However, performance often degrades in noisy or highly contextual real-world clinical environments, and hallucinations, inconsistent outputs, and limited robustness remain significant barriers to clinical deployment, [11], [12] and [9].

Parallel work has explored the use of LLMs for patient-trial matching, ranging from zero-shot prompting approaches to multi-stage retrieval and re-ranking pipelines, [12], [4] and [8]. These systems typically represent patients or trials as queries, simplify eligibility criteria into binary questions, [8], [3] and [7], or classify eligibility using categorical or percentage-based scoring, [5], [4] and [3]. Results consistently show that LLMs can match or surpass expert performance, particularly when enhanced with ontologies, retrieval-augmented architectures, or task-aligned fine-tuning, [7], [8], [4] and [5]. Open-source

models have also demonstrated competitive performance relative to proprietary systems, while offering substantial reductions in screening time and computational cost, [5], [3] and [4]. Nonetheless, these approaches face limitations, including high false-negative rates caused by non-standard abbreviations and ontology gaps, reliance on synthetic datasets that may introduce bias, sensitivity to prompt design, [5], [2] and [12], and challenges related to privacy and regulatory compliance, [4], [12] and [6].

More recent studies have begun to explore hybrid pipelines that integrate information extraction and patient-trial matching within unified frameworks, [4] and [3]. These systems typically follow an extraction-first paradigm, [7], or employ multi-agent LLM setups that combine extraction and reasoning, [5] and [4]. However, most hybrid approaches rely on end-to-end probabilistic LLM reasoning or assume access to structured electronic health records, [12] and [4], blurring the distinction between deterministic rule-based evaluation and LLM-assisted interpretation. As a result, few systems explicitly separate the interpretation of unstructured clinical text from the transparent, reproducible evaluation of eligibility criteria.

Taken together, the literature highlights several gaps that remain insufficiently addressed. Clinical information extraction and patient-trial matching are often treated as independent tasks, [11] and [9], and evaluations frequently rely on curated or synthetic datasets that do not reflect the complexity of real-world oncology workflows, [6], [4], [12] and [3]. Eligibility assessment is commonly performed through probabilistic LLM reasoning, limiting transparency, auditability, and reproducibility, [4] and [12]. Only a small number of studies propose integrated workflows that combine clinician-validated extraction, structured patient representation, formalization of eligibility criteria into machine-readable rules, [3], and hybrid eligibility assessment mechanisms capable of combining deterministic evaluation with LLM-assisted reasoning when necessary. This gap motivates the need for approaches that bridge the flexibility of LLMs with the transparency of rule-based systems while operating directly on real-world unstructured oncology data.

SYSTEM DESIGN AND ARCHITECTURE

As described above the present approach aims to reduce work-overload and screen time of oncologist experts by leveraging the use of LLMs to extract hyper-relevant parameters of free-text clinical diaries forming this way validated structured patient profiles, that in conjunction with the criteria extraction and conversion, will allow for a faster, automatic and semi-deterministic matching of potentially eligible patients for given clinical trials. Given the non-deterministic nature of LLMs and the critical clinical context, all automated steps require expert validation by the clinician.

At a methodological level, the system pipeline is divided into interconnected modules phases:

- Phase 1: Hyper-relevant Parameter Extraction;
- Phase 2: Extraction and Conversion of Clinical Criteria;
- Phase 3: Semi-Deterministic Trial-Patients Matching.

ARCHITECTURE

Flow

The proposed architecture follows a sequential workflow in which each phase produces artifacts required by subsequent stages. The Hyper-Relevant Parameter Extraction phase generates clinician-validated structured patient profiles from free-text clinical diaries. In parallel, the Extraction and Conversion of Clinical Criteria phase processes clinical trial documentation to produce validated eligibility criteria and their corresponding machine-readable logical representations.

The communication between phases is performed through the system database, where all intermediate artifacts are persisted after clinician validation. This design ensures traceability, reproducibility, and allows experts to review and modify the outputs of each phase before they are consumed by downstream components.

Once both structured patient profiles and validated eligibility rules are available, the Semi-Deterministic Trial-Patient Matching phase can be executed. This final phase retrieves the artifacts produced by the previous stages and evaluates patient eligibility for a selected clinical trial, producing a ranked list of potentially eligible patients together with criterion-level assessment results.

Hyper-Relevant Parameter Extraction

The objective of the first phase is to transform unstructured free-text clinical information into normalized and structured patient data. The input consists of clinical diaries written in free-text format, which are selected by the clinician through the system interface.

The extraction process comprises the following steps:

- 1) Text Pre-processing: A clinical diary uploaded to the system is selected and pre-processed. During this stage, irrelevant sections such as headers and footers are removed to reduce unnecessary context and computational overhead.
- 2) Information Extraction and Normalization: The pre-processed text is provided to the LLM together with a task-specific prompt. The model is instructed to identify and extract clinically relevant parameters according to a predefined schema. During the same step, diagnosis normalization is performed using a stakeholder-provided mapping to ensure consistency across patient records.
- 3) Validation and Structured Output Generation: The extracted information is presented to the clinician for expert validation and potential correction. Once approved, the validated patient profile is persisted in the database

and becomes available for subsequent phases of the workflow.

A formalization of this workflow can be seen on the Figure 1 below.

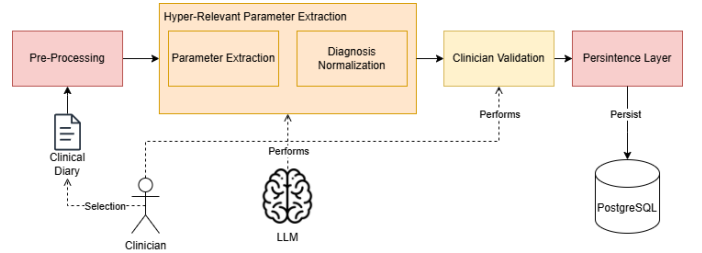


Fig. 1. Hyper-Relevant Parameter Extraction

Upon completion of this phase, the system produces a structured repository of validated patient profiles containing the key clinical parameters required for eligibility assessment.

Patient Profile Schema: The extraction schema follows a JSON structure composed of the following fields:

- Age: Patient age or date of birth, depending on the information available in the clinical diary;
- Gender: Patient gender;
- ECOG-PS: ECOG Performance Status score (0–5), describing the patient’s functional capacity;
- Diagnosis: Primary diagnosis, typically corresponding to the patient’s cancer type or major clinical condition;
- Diagnosis Date: Date on which the diagnosis was formally established;
- Molecular Status: Relevant molecular and biomarker information;
- Stage: Disease staging information;
- Pathology Group: Clinical category grouping diagnoses according to the organ or system of origin;
- Treatments: List of administered treatments:
 - Name: Treatment name;
 - Start Date: Date on which the treatment started;
 - End Date: Date on which the treatment ended;
- Control: Information related to disease control, metastases, comorbidities, and other relevant clinical follow-up observations.

All schema fields are provided to the LLM as target extraction parameters. However, as clinical diaries may not contain all required information, the model is allowed to return null values for unavailable fields. Since all extracted information undergoes clinician validation before persistence, missing or uncertain values can be reviewed and corrected when necessary.

Error Handling and Robustness: To improve robustness, several error scenarios are explicitly handled during extraction:

- Invalid JSON Structure: If the LLM generates explanatory text together with the JSON output, the system

attempts to isolate and recover the valid JSON structure. If recovery is unsuccessful, the extraction request is automatically repeated.

- **Missing JSON Output:** If no valid JSON structure is generated, the extraction request is repeated.
- **Truncated Responses:** If the generated output is incomplete due to context or generation limits, the extraction request is repeated.

The objective of these recovery mechanisms is to ensure that a valid and structurally consistent JSON representation is obtained before the extracted information is presented for clinician validation.

Produced Artifacts: Upon completion of this phase, a single primary artifact is generated and made available through the system interface:

- **Structured Patient Profile:** A clinician-validated representation of the patient containing the extracted clinical parameters required for subsequent eligibility assessment.

The resulting structured profiles provide a normalized representation of patient information that can be consistently queried and evaluated during the matching process, while maintaining traceability through clinician validation.

Extraction and Conversion of Clinical Criteria

The objective of the second phase is to transform clinical trial eligibility requirements into machine-interpretable representations that can later be used during patient eligibility assessment. The input to this phase consists of clinical trial documentation uploaded, managed, and selected through the system interface.

The process comprises the following steps:

- 1) **Text Pre-processing:** The clinician selects a clinical trial document, which undergoes pre-processing to remove sections that are not relevant to eligibility assessment. Similar to the clinical diary pipeline, this step reduces unnecessary context and computational overhead.
- 2) **Trial Structure Identification:** Before extracting eligibility criteria, the LLM analyses the clinical trial document to identify its internal structure, namely the existence of trial cohorts. This information is subsequently used during criteria extraction and patient eligibility assessment.
- 3) **Criteria Extraction:** The LLM receives the processed trial documentation together with the identified trial structure and extracts inclusion and exclusion criteria. Whenever possible, criteria are extracted individually and associated with their corresponding cohort. However, some eligibility requirements must remain grouped to preserve their original clinical meaning. The system therefore supports both simple and compound criteria.
- 4) **Extracted Criteria Validation and Persistence:** The extracted criteria are presented to the clinician for expert validation. Once approved, they are persisted in the

database, ensuring traceability and providing the input required for the next step of the pipeline.

- 5) **Criteria Conversion:** Following validation, each criterion is translated into a structured logical representation. The objective of this step is to maximize the use of deterministic evaluation and reduce reliance on probabilistic LLM reasoning during eligibility assessment.
- 6) **Logical Representation Validation and Persistence:** The generated logical representations are presented to the clinician for validation to ensure that they accurately reflect the meaning of the original eligibility criteria. Once approved, they are stored in the database and become available for patient eligibility assessment.

A formalization of this workflow can be seen on the Figure 2 below.

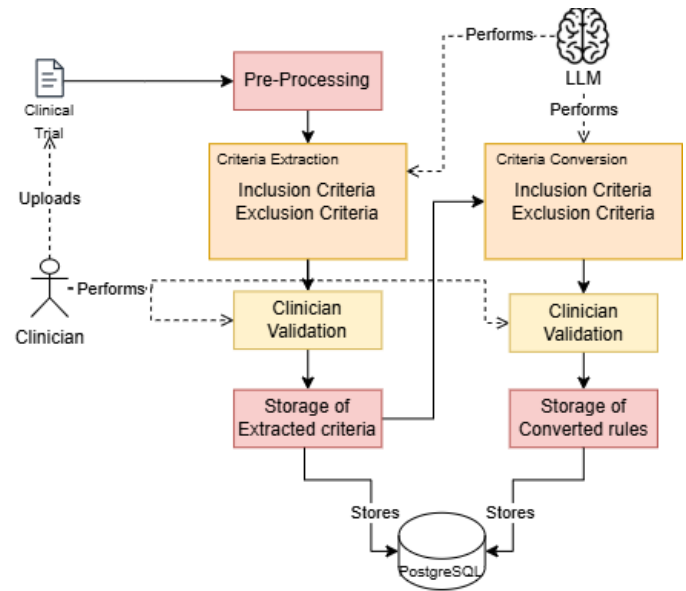


Fig. 2. Extraction and Conversion of Clinical Criteria Workflow

Initially, criteria extraction and logical conversion were designed as a single task. However, previous studies have shown that decomposing complex tasks into simpler sequential steps often leads to improved performance and reliability. For this reason, the structure identification, extraction and conversion stages were implemented as separate phases.

Conversion of Criteria: The LLM is provided with a prompt that guides it to perform the best possible conversion for the system. It tries at all costs, while maintaining its meaning to convert the criterion into a valid field available in the patient profile schema, or in the laboratory values schema. Only if it is incapable of such, the LLM is instructed to generate a general field for it. These incompatible fields will be discussed forward in the sections and

Complex Criteria Handling: The identification of simple and compound criteria is performed by the LLM. The model receives the complete textual description of each eligibility criterion and determines whether it can be safely decomposed

into independent criteria without altering its clinical meaning. If separation preserves the original semantics, the criterion is divided into multiple individual criteria. Otherwise, it is maintained as a compound criterion and processed as a single logical unit.

Logical Representation of Eligibility Criteria: To enable machine-interpretable and partially deterministic eligibility assessment, each validated criterion is converted into a JSON-based logical representation.

Simple criteria are represented as atomic rules composed of a field, an operator, and a value. For example:

```
{
  "field": "age",
  "operator": ">=",
  "value": "18"
}
```

More complex criteria involving multiple conditions are represented through Boolean operators such as AND and OR, where each operator groups a set of sub-conditions. For example:

```
{
  "operator": "AND",
  "conditions": [
    {
      "field": "ecog_ps",
      "operator": "<=",
      "value": "1"
    },
    {
      "field": "diagnosis",
      "operator": "=",
      "value": "NSCLC"
    }
  ]
}
```

In some cases, a criterion cannot be automatically mapped to the predefined logical schema. In these situations, the following representation is generated, allowing the clinician to manually define the corresponding logical rule:

```
{
  "unmapped": true,
  "source_text": "unmapped part of the criterion"
}
```

Finally, all converted criteria are grouped according to their category, resulting in the following structure:

```
{
  "inclusion_criteria": [
    {
```

```
      "id": integer,
      "text": string,
      "logic": object
    }
  ],
  "exclusion_criteria": [
    {
      "id": integer,
      "text": string,
      "logic": object
    }
  ]
}
```

Produced Artifacts: Upon completion of this phase, two primary artifacts are generated and made available through the system interface:

- **Clinical Trial Criteria:** A validated list of individually extracted eligibility criteria, organized into inclusion and exclusion categories;
- **Logical Rules:** The corresponding machine-readable logical representations of those criteria, also organized into inclusion and exclusion categories.

This representation allows the system to uniformly encode both simple and compound eligibility requirements. Criteria that reference fields available in the patient profile schema can subsequently be evaluated deterministically, while criteria that cannot be represented through the schema remain available for LLM-assisted assessment during the matching phase. This hybrid approach increases transparency and auditability while preserving the flexibility required to handle complex clinical requirements.

Semi-Deterministic Trial-Patient Matching

The objective of the final phase is to identify potentially eligible patients for a given clinical trial through a semi-deterministic eligibility assessment process. This phase consumes the structured patient profiles generated during the first phase and the validated logical representations of eligibility criteria produced during the second phase.

The matching process comprises the following steps:

- 1) **Trial Selection and Patient Retrieval:** A clinician selects the clinical trial to be assessed. Based on the metadata defined during trial registration, a subset of patient profiles is retrieved from the database. Patient selection is initially restricted to the pathology group associated with the clinical trial, reducing the number of comparisons and LLM calls required during eligibility assessment.
- 2) **Eligibility Assessment:** Each patient is evaluated against every validated eligibility criterion. Two assessment paths are available:
 - **Deterministic Assessment:** If the criterion references a field available in the patient profile schema,

the corresponding logical rule is evaluated directly against the structured patient data.

- **LLM-Assisted Assessment:** If the criterion references information that is not represented within the structured schema, the LLM receives the patient profile, clinical diary, laboratory results, and the corresponding criterion to determine eligibility. In addition to the assessment result, the model provides a justification to support expert review.

Although different evaluation mechanisms are employed, both paths produce the same output: a criterion-level eligibility assessment that is subsequently used during final eligibility determination.

- 3) **Assessment Validation and Persistence:** All criterion-level assessments are presented to the clinician for review. Experts may modify any automatically generated decision, regardless of whether it originated from deterministic evaluation or LLM-assisted reasoning. Following validation, all criterion assessments, manual corrections, and final eligibility decisions are persisted in the database to ensure traceability and auditability.

A formalization of this workflow can be seen on the Figure 3 below.

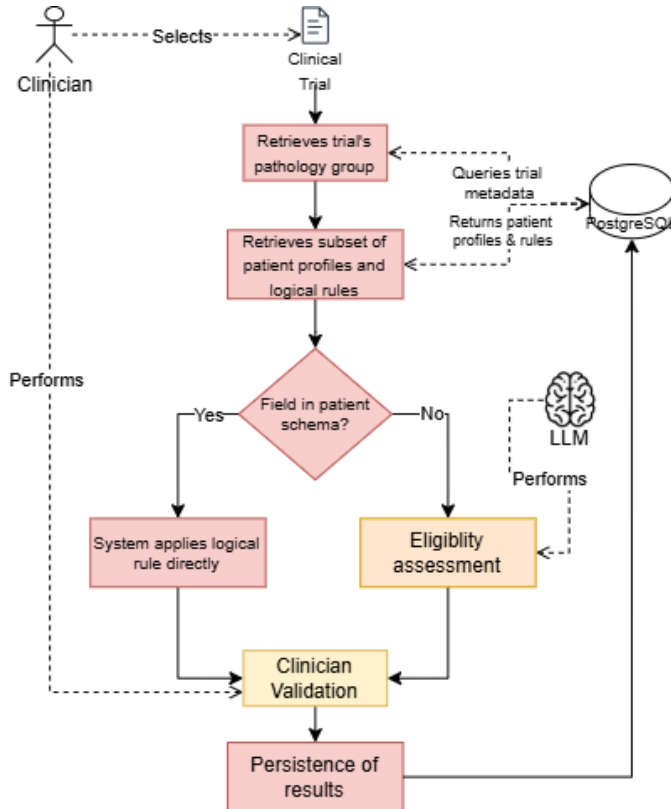


Fig. 3. Semi-Deterministic Trial-Patient Matching

Matching Strategy: The proposed matching approach is semi-deterministic. Whenever a criterion references information available in the structured patient profile schema, eli-

gibility is assessed through deterministic logical evaluation. Conversely, criteria that cannot be directly represented within the schema are evaluated through LLM-assisted reasoning.

This design maximizes the use of transparent and reproducible rule-based assessment while preserving the flexibility required to evaluate complex clinical requirements that cannot be formally encoded.

Eligibility Aggregation: The final patient eligibility decision is computed through a strict rule-based aggregation of all criterion-level assessments. Criteria are divided into inclusion and exclusion criteria and are evaluated independently before aggregation.

A patient is considered eligible if:

- All inclusion criteria are satisfied; and
- No exclusion criterion is triggered.

Formally:

Eligible = (All Inclusion Criteria Satisfied)
AND
(No Exclusion Criteria Triggered)

No weighting, ranking, or probabilistic scoring mechanism is employed. The final decision relies exclusively on logical consistency across all eligibility criteria.

For clinical trials containing multiple cohorts, eligibility is evaluated at both the trial and cohort levels. A patient is considered eligible for the trial if the general eligibility requirements are satisfied and at least one cohort-specific eligibility assessment returns an eligible result.

Formally, a patient is considered eligible if:

- The general trial criteria are satisfied; and
- At least one cohort satisfies its respective inclusion and exclusion requirements.

Inconclusive Assessments: Certain criteria may be impossible to evaluate reliably. This can occur due to missing clinical information, unavailable laboratory values, unit mismatches, parsing failures, ambiguous criterion definitions, or criteria that cannot be resolved through either deterministic evaluation or LLM-assisted assessment.

In these situations, the corresponding criterion is classified as inconclusive.

To ensure a conservative clinical decision-support strategy:

- Inconclusive inclusion criteria are treated as not satisfied;
- Inconclusive exclusion criteria are treated as not triggered.

This approach prioritizes patient safety by avoiding positive eligibility decisions when essential information is unavailable or uncertain.

Produced Artifacts: Upon completion of this phase, two primary artifacts are generated and made available through the system interface:

- **Criterion Assessments:** Individual eligibility decisions for each criterion, together with supporting justifications whenever LLM-assisted assessment is employed;
- **Eligibility Results:** A list of patients classified according to their eligibility status for the selected clinical trial and, when applicable, for individual cohorts.

The resulting assessments provide full traceability from the final eligibility decision down to the individual criterion evaluations, allowing clinicians to review, validate, and audit every step of the matching process.

MATERIALS AND METHODS

Documents as Inputs

This section describes the two document types processed by the proposed system and the abstractions adopted to model them within the pipeline.

The proposed framework operates on two primary document sources: clinical diaries and clinical trial protocols. Although both are written in free-text format, they differ substantially in their structure, purpose, and degree of standardization. Understanding these characteristics is essential for the design of the extraction and matching pipelines.

Clinical Trials: Clinical trial protocols are predominantly narrative documents describing study objectives, methodology, treatment arms, safety procedures, and participant selection requirements. Despite variations across studies, trial protocols generally follow a well-defined organizational structure.

Among all protocol sections, eligibility criteria constitute the only component directly involved in patient selection. Inclusion and exclusion criteria are expressed in natural language but fundamentally represent constraints over patient attributes, such as demographic characteristics, diagnoses, biomarkers, treatment history, laboratory values, or performance status.

However, eligibility structures vary considerably between trials. Some studies define a single set of eligibility requirements, while others contain multiple cohorts, each with its own inclusion and exclusion criteria. Furthermore, eligibility requirements may range from simple atomic conditions to highly complex compound criteria involving multiple logical dependencies.

To enable automated processing, the proposed system abstracts clinical trial documents into two components: trial structure and eligibility criteria. This separation allows descriptive protocol information to remain independent from the patient selection logic. Consequently, LLM-based extraction can focus exclusively on eligibility requirements, which are subsequently converted into machine-readable logical representations suitable for automated evaluation.

All clinical trials used throughout this study were obtained in English.

Clinical Diaries: Clinical diaries represent the primary source of patient information used by the system. Unlike clinical trial protocols, these documents exhibit substantial variability in writing style, organization, terminology, and level of detail. Such variability arises from differences between institutions, clinical specialties, individual physicians, and documentation practices.

The design of the extraction pipeline was guided by a structured template provided by the clinical stakeholder and derived from four representative oncology clinical diaries routinely used within the Oncology Department of Coimbra University Hospital Centre (HUC). Rather than defining a fixed document structure, these diaries were analysed to identify recurring clinical concepts and information categories required for patient eligibility assessment.

From this analysis, a patient profile schema was defined, containing the hyper-relevant clinical parameters considered essential for trial matching. These parameters serve as the target representation of the extraction pipeline and provide a consistent structured format regardless of how information is expressed in the original diary.

To avoid overfitting the system to a specific documentation template, the abstraction process focused on preserving semantic information rather than document layout or wording. Structural elements, section ordering, formatting conventions, and stylistic patterns were intentionally disregarded whenever they were not clinically relevant. Only information categories consistently required for eligibility assessment, such as diagnosis, staging, treatments, biomarkers, ECOG performance status, and disease evolution, were retained in the final schema.

This abstraction enables the system to process heterogeneous clinical documentation while producing a standardized representation suitable for automated eligibility assessment. Although the proposed schema was derived from documents used within a specific oncology department, it was designed to capture clinically meaningful concepts rather than institution-specific formatting conventions.

All clinical diaries used throughout this study were written in European Portuguese.

Datasets

This section describes the datasets used to evaluate the components of the proposed system, including synthetic oncology clinical diaries, clinical trial documentation, and manually curated gold standards. A quantitative summary of these normalized datasets is provided in Table I. All data used in this study were fully anonymized; no identifiable patient information was used, and every synthetic diary contains no real clinical content.

Synthetic Oncology Clinical Diaries: The system evaluation was conducted using a dataset of 30 synthetic free-text oncology clinical diaries. These diaries were generated using the open-source `gpt-oss-120b` model, selected for its high generation capacity, and were grounded on four representative clinical diaries provided by the clinical stakeholder to serve as references for writing style, structure, and clinical content. The dataset size balances clinical diversity with the feasibility of manual annotation.

To ensure representativeness, 30 unique patient profiles were first created, each describing a distinct oncological case. The generation pipeline consisted of two sequential stages designed to reproduce the variability observed in real-world clinical documentation:

- 1) **Idealized Generation:** The LLM generated a clinically coherent and internally consistent Portuguese oncology note from the patient profile, maintaining alignment between diagnosis, staging, treatments, imaging findings, ECOG performance status, comorbidities, and disease evolution.
- 2) **Noise and Artifact Insertion:** To resemble routine hospital records, the generated diary was transformed by introducing realistic imperfections while preserving complete clinical correctness. For each diary, two generation factors were randomly assigned: a *length mode* (short, medium, or long) and a *style mode* (narrative-dominant, telegraphic hospital style, exam-and-imaging focused, toxicity focused, or psychosocial emphasis). The model introduced variations in sentence length, irregular formatting, heterogeneous abbreviation usage, minor spelling imperfections, structural asymmetries (e.g., missing headers), and administrative workflow notes.

To guide this process, the prompts incorporated examples of authentic clinical documentation exclusively for stylistic learning, with explicit instructions not to reproduce original clinical content. Following generation, an experienced oncologist reviewed and validated all diaries for clinical plausibility and consistency.

After normalization (removing headers, footers, and irrelevant sections), the clinical diary dataset contained a total of 64,953 tokens across the 30 documents, averaging 2,165 tokens per diary.

Clinical Trials Dataset: A dataset of 30 clinical trials was assembled from *ClinicalTrials.gov* to evaluate the criteria extraction, conversion, and patient-trial matching pipelines, seeking to maximize clinical diversity and minimize selection bias.

The trial collection covers a broad range of tumor types (including breast, lung, pancreatic, colorectal, melanoma, among others), disease stages, molecular profiles, and therapeutic settings. The selected trials encompass different designs (with and without cohort structures) and exhibit substantial variability in

eligibility complexity, ranging from simple demographic or diagnostic constraints to highly specialized molecular biomarkers and prior treatment histories.

After normalization, the clinical trial dataset contained a total of 120,786 tokens across the 30 protocols, averaging 4,026 tokens per trial, reflecting the complexity of oncology requirements. The eligibility criteria corpus comprises a total of 892 individual criteria (480 inclusion and 412 exclusion criteria), averaging approximately 30 eligibility criteria per trial.

Manually Curated Gold Standards: All gold standards were manually annotated to provide reliable reference data for system evaluation across four core tasks:

- **Hyper-Relevant Parameter Extraction:** All fields defined in the patient profile schema were manually identified and annotated within the 30 clinical diaries to enable automated, deterministic comparisons with the LLM outputs.
- **Trial Structure Identification:** The organization of eligibility requirements and the presence of cohorts were manually mapped following the system’s formal representation.
- **Criteria Extraction:** All 892 inclusion and exclusion criteria were manually extracted as individual requirements to evaluate the completeness of the automated extraction.
- **Criteria Conversion:** Expected logical representations were defined according to the system’s conversion schema. Criteria mapping to predefined patient profile fields were annotated with logical rules for deterministic evaluation. Criteria depending on unstructured clinical context were assessed qualitatively to ensure semantic consistency.

TABLE I
QUANTITATIVE SUMMARY OF THE NORMALIZED DATASETS USED IN THIS STUDY

Dataset Component	Total	Average per Document
Clinical Diaries (tokens)	64,953	2,165
Clinical Trials (tokens)	120,786	4,026
Inclusion Criteria	480	16
Exclusion Criteria	412	14
Total Criteria	892	30

Proposed Method

The proposed method aims to support clinical trial recruitment by combining clinical information extraction and patient-trial matching within a single LLM-assisted framework. Rather than relying exclusively on probabilistic reasoning, the approach seeks to maximize deterministic processing whenever possible by transforming both patient information and trial eligibility criteria into structured representations that can be automatically compared.

The method is composed of three sequential tasks. First, relevant clinical information is extracted from free-text oncology

clinical diaries and transformed into structured patient profiles. Second, eligibility criteria are extracted from clinical trial documentation and converted into machine-interpretable logical representations. Finally, patient profiles and trial criteria are combined through a semi-deterministic matching process to identify potentially eligible patients.

The central premise of the proposed approach is that eligibility assessment can be significantly improved by separating complex reasoning into multiple specialized tasks. Instead of asking a model to directly determine patient eligibility from raw documents, the system decomposes the problem into information extraction, criteria formalization, and eligibility assessment stages. This decomposition aims to improve interpretability, traceability, and clinician oversight while reducing reliance on end-to-end LLM reasoning.

The overall system was designed with real clinical environments in mind and was iteratively discussed with domain experts to ensure practical applicability. As a consequence, every LLM-assisted step was built to guarantee interpretability and auditability by allowing direct comparison between outputs and their source documents, while storing and clearly labeling all intermediate results. Throughout the pipeline, the clinician remains the final approver and editor of all outputs, ensuring clinical safety while leveraging LLMs to accelerate the workflow.

A secondary objective of this study is to evaluate whether relatively small, non-proprietary open-source language models can achieve satisfactory performance across the different stages of the pipeline. Demonstrating competitive performance with such models would increase the feasibility of deploying the proposed framework in environments with limited computational resources and stricter privacy requirements.

The proposed method is evaluated through the following tasks, each task is evaluated independently to isolate error sources and quantify the contribution of each module:

- Extraction and normalization of hyper-relevant patient information from free-text clinical diaries;
- Identification of clinical trial eligibility structures, (cohort detection);
- Extraction of individual inclusion and exclusion criteria;
- Conversion of eligibility criteria into machine-interpretable logical representations;
- Semi-deterministic patient-trial matching through the combination of deterministic and LLM-assisted assessments.

The study is guided by the following research hypotheses:

- H1: Large Language Models can accurately extract and normalize hyper-relevant clinical information from heterogeneous oncology clinical diaries while adhering to a predefined structured schema.
- H2: Large Language Models can accurately identify, extract, and formalize clinical trial eligibility criteria while

preserving their original clinical meaning and logical relationships.

- H3: A hybrid eligibility assessment strategy combining deterministic rule evaluation and LLM-assisted reasoning can effectively support patient-trial matching while maintaining interpretability and clinician control.
- H4: Small open-source language models can achieve performance levels sufficient to support the proposed workflow across the evaluated tasks.

Experimental Setup

As previously described, the proposed framework consists of three LLM-assisted pipelines:

- Hyper-Relevant Parameter Extraction;
- Extraction and Conversion of Clinical Criteria;
- Semi-Deterministic Patient-Trial Matching.

The experimental evaluation was designed to assess the performance of the LLM-dependent components of each pipeline independently. Evaluating each stage separately allows the identification of potential error sources and provides a clearer understanding of the contribution of each component to the overall system performance.

The experiments aim to evaluate both the effectiveness and robustness of the proposed approach, as well as its suitability for integration into a semi-automated clinical decision support workflow. Particular emphasis is placed on accuracy, reliability, interpretability, and the ability of the system to operate under clinician supervision.

Evaluation Scenarios: To evaluate the different stages of the proposed framework, two primary experimental scenarios were defined.

Scenario 1 – Clinical Diary Processing

This scenario evaluates the Hyper-Relevant Parameter Extraction pipeline. Each model receives an individual clinical diary and is tasked with extracting and normalizing the information required to populate the predefined patient profile schema.

The objective is to assess the ability of the models to identify clinically relevant information from heterogeneous free-text oncology documentation while generating outputs that comply with the required structured representation.

Scenario 2 – Clinical Trial Processing

This scenario evaluates the Extraction and Conversion of Clinical Criteria pipeline. Each model receives an individual clinical trial protocol and performs the following tasks:

- Identification of the trial eligibility structure, including cohort detection when applicable;
- Extraction of inclusion and exclusion criteria;
- Conversion of extracted criteria into machine-interpretable logical representations.

The objective is to assess whether the models can preserve the clinical meaning of eligibility requirements while transforming them into structured representations suitable for automated evaluation.

Scenario 3 - Patient-Trial Matching This scenario evaluates the LLM-assisted component of the Semi-Deterministic Patient-Trial Matching pipeline. In cases where eligibility criteria cannot be directly evaluated through deterministic logical rules, either due to incomplete formalization or the presence of schema-independent requirements, the model receives the structured patient profile, the original clinical diary, relevant laboratory information, and the eligibility criterion represented as a logical rule. The model is then tasked with determining whether the available patient information satisfies the given criterion and providing a clinically grounded justification for the decision.

The objective is to assess the ability of the models to perform reliable eligibility assessments when operating under constrained and explicit clinical context, as well as the quality and transparency of the generated clinical justifications.

All models are evaluated under identical experimental conditions to ensure fair and reproducible comparisons.

To further improve reproducibility, prompts are version-controlled and preserved throughout the experiments, and all model configuration parameters remain fixed across evaluation runs.

Evaluation Metrics: Different evaluation metrics are employed depending on the task under analysis.

Hyper-Relevant Parameter Extraction

The extraction of structured patient information is evaluated using:

- Field-level Accuracy: proportion of correctly extracted fields relative to the manually annotated gold standard;
- Missing Field Rate: proportion of schema fields returned as null or omitted;
- Schema Compliance Rate: percentage of outputs that fully comply with the predefined JSON schema.

Trial Structure Identification

The identification of clinical trial structure is evaluated through:

- Cohort Detection Accuracy: correctness of identifying whether cohort structures are present;
- Structure Description Correctness: expert assessment of the generated trial structure representation.

Eligibility Criteria Extraction

The extraction of inclusion and exclusion criteria is evaluated through:

- Criteria Extraction Correctness: manual and expert assessment of whether extracted criteria faithfully represent

the original trial requirements.

Eligibility Criteria Conversion

The conversion of eligibility criteria into logical rules is evaluated through:

- Logical Representation Correctness: Manual assessment indicating whether the generated logical representation accurately reflects the clinical meaning of the original criterion.

Patient-Trial Matching

The semi-deterministic matching process is evaluated through:

- Eligibility Assessment Accuracy: correctness of the generated eligibility decision relative to the reference documentation of the patient (clinical diary, laboratory values, and patient profile);
- Justification Quality: expert evaluation of the clinical coherence and usefulness of generated explanations using a five-point Likert scale.

Evaluated Models: The experiments compare multiple open-source Large Language Models representing different model families and architectural approaches:

- Qwen3 8B Instruct;
- Llama 3.1 8B Instruct;
- DeepSeek-R1 8B.

These models were selected because they provide a representative sample of state-of-the-art open-source LLMs that can be deployed with relatively modest computational resources.

All models are evaluated using deterministic inference settings to minimize output variability and maximize reproducibility. No model fine-tuning is performed, and all experiments are conducted under a zero-shot setting. This configuration reflects the intended deployment scenario of the proposed framework, where task-specific training data may not be available and rapid adaptation to new clinical contexts is desirable.

Each experiment was executed once under deterministic inference settings, on the datasets of each 30 clinical diaries and 30 clinical trials.

RESULTS

As mentioned before the selected language models were evaluated in 5 tasks: Hyper-Relevant Parameter Extraction; Trial Structure Identification; Eligibility Criteria Extraction; Eligibility Criteria Conversion; and Patient-Trial Matching.

Hyper-Relevant Parameter Extraction

The first experimental task evaluated the capability of the LLMs to extract clinically relevant information from free-text oncology clinical diaries while adhering to a predefined JSON

schema. The evaluation focused on three main aspects: field-level extraction accuracy, schema compliance, and missing field rate. Field-level accuracy was computed by manually comparing the generated patient profiles against the manually annotated gold standards. Two values are reported: strict accuracy, considering only fully correct extractions, and a partial accuracy score, which also accounts for partially correct extractions.

As shown in Table II, all evaluated models were able to consistently follow the predefined output structure, achieving 100% schema compliance. This demonstrates that the evaluated models were capable of generating structurally valid outputs according to the required JSON representation, regardless of their differences in extraction performance.

Regarding extraction accuracy, Qwen3-8B achieved the highest overall performance, obtaining a strict accuracy of 78.3% and a partial accuracy score of 83.3%. DeepSeek-R1-8B followed with 74.3% strict accuracy and 80.5% partial accuracy. Llama3.1-8B presented the lowest performance, achieving 62.3% strict accuracy and 70.3% partial accuracy, with a higher number of incorrect and partially extracted fields. These results indicate that although all models were capable of producing valid structured outputs, their ability to correctly identify and populate clinical information varied considerably. Figure 6 illustrates the comparison between strict and partial accuracy across models.

The missing field rate followed a similar trend. Qwen3-8B and DeepSeek-R1-8B presented lower rates of missing information (11.6% and 13.3%, respectively), whereas Llama3.1-8B showed a considerably higher missing field rate of 21.7%. This suggests that weaker models had more difficulty identifying whether relevant clinical information was present within the diary and subsequently extracting it into the required schema fields.

Table III provides a more detailed field-level analysis across the patient profile schema. Some fields were consistently extracted with high accuracy across all models. Gender and ECOG-PS achieved 100% accuracy for every evaluated model, indicating that these parameters are generally explicit and unambiguous within oncology clinical documentation. Figure 4 further illustrates the performance distribution across schema fields.

Conversely, more complex fields requiring interpretation, normalization, or structured reasoning presented lower performance. Diagnosis extraction represented one of the most challenging fields, with DeepSeek-R1-8B achieving the highest accuracy at 46.7%. However, manual inspection revealed that most errors were not related to failure in identifying the diagnosis itself, but rather to difficulties in applying the required normalization mapping. In several cases, models extracted the correct clinical diagnosis but failed to convert it into the expected normalized representation.

Treatment extraction was another challenging task due to its

hierarchical structure, requiring the identification of multiple sub-fields, including treatment name, start date, and end date. Qwen3-8B achieved the best performance in this field, with 70.0% strict accuracy and an 85.0% partial accuracy score, while Llama3.1-8B achieved substantially lower results.

Disease staging also proved challenging, particularly due to the need to preserve the complete staging representation. A recurrent error pattern was the extraction of only the final stage classification (e.g., "IVB") instead of the complete TNM representation (e.g., "cT3N2M1b (IVB)"). Although the clinically relevant information was partially captured, these outputs were considered incomplete according to the defined schema.

Date-related fields, particularly diagnosis dates and treatment periods, also represented a source of errors across models. These errors consisted mainly of incorrect extraction, incomplete temporal information, or returning null values when dates were present in the clinical diary.

Overall, the results demonstrate that lightweight LLMs are capable of performing structured clinical information extraction from heterogeneous oncology documentation. However, performance is strongly influenced by the complexity of the target field, with explicit clinical attributes being extracted reliably, while fields requiring normalization, temporal reasoning, or hierarchical interpretation remain more challenging. Figure 5 summarises the schema compliance rate across all evaluated models.

TABLE II
GLOBAL PERFORMANCE OF THE EVALUATED MODELS (HYPER-RELEVANT PARAMETER EXTRACTION)

Models	Total	Correct	Partial	Wrong	Acc/Partial (%)	Schema (%)	Missing (%)
Llama3.1-8b	300	187	48	65	62.3/70.3	100.0	21.7
Deepseek-r1-8b	300	223	37	40	74.3/80.5	100.0	13.3
Qwen3-8b	300	235	30	35	78.3/83.3	100.0	11.6

TABLE III
FIELD-LEVEL PERFORMANCE ACROSS SCHEMA COMPONENTS

Schema Fields	Llama3.1-8b	Deepseek-r1-8b	Qwen3-8b
age_or_birthdate	46.6/46.6	100.0/100.0	100.0/100.0
gender	100.0/100.0	100.0/100.0	100.0/100.0
ecog_ps	100.0/100.0	100.0/100.0	100.0/100.0
diagnosis	36.7/36.7	46.7/46.7	43.3/43.3
diagnosis_date	56.7/56.7	63.3/63.3	76.7/76.7
molecular_status	60/71.7	66.7/73.3	63.3/71.7
stage	23.3/61.7	23.3/61.7	46.7/73.3
pathology_group	63.3/63.3	90.0/90.0	83.3/83.3
treatments	36.7/66.7	56.7/73.3	70.0/85.0

Trial Structure Identification

The second experimental task evaluated the capability of the LLMs to identify and represent the internal structure of clinical trials according to a predefined JSON schema. This step is required for subsequent eligibility assessment, as cohort-specific requirements may define different patient populations within the same clinical trial.

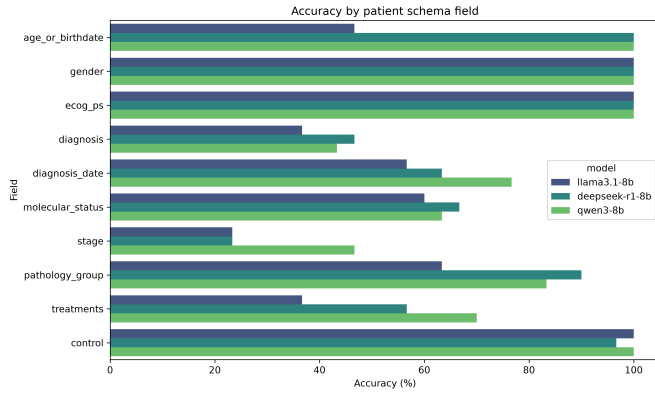


Fig. 4. Hyper-Relevant Parameter Extraction - Accuracy by schema field

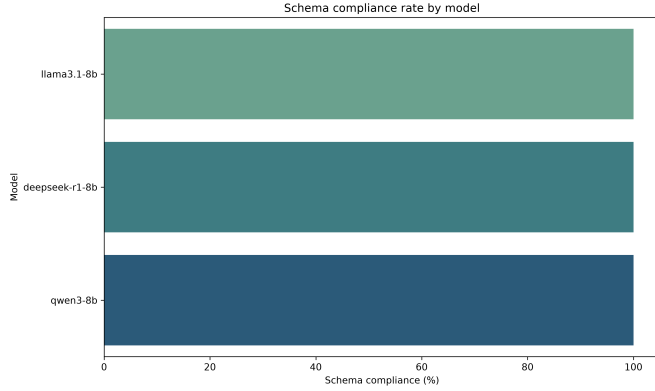


Fig. 5. Hyper-Relevant Parameter Extraction - Schema compliance rate by model

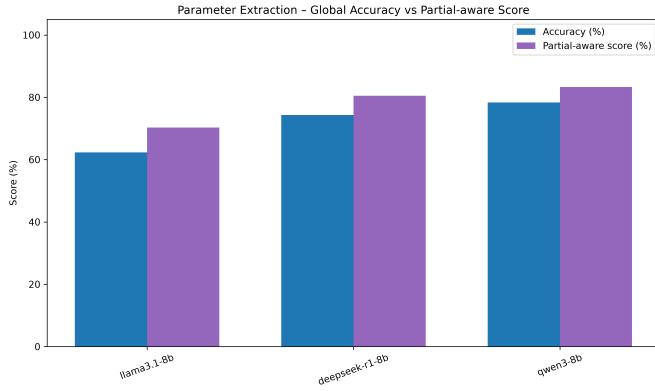


Fig. 6. Hyper-Relevant Parameter Extraction - strict vs partial accuracy

The evaluation considered two aspects: (i) identification of whether formal cohort structures were present, and (ii) correctness of the generated cohort representation. The generated outputs were compared against manually annotated gold standards. Strict accuracy, considering only fully correct outputs, and partial accuracy, accounting for partially correct representations, were reported.

As shown in Table IV, substantial differences were observed

between models. Deepseek-r1:8b achieved the best performance, correctly identifying 58 out of 60 evaluated fields, corresponding to 96.7% strict accuracy. In contrast, Llama3.1-8b achieved considerably lower performance (25.0%), mainly due to difficulties distinguishing formal trial cohorts from descriptive patient groups. Qwen3-8b obtained intermediate results, achieving 60.0% strict accuracy and 75.0% partial accuracy.

TABLE IV
GLOBAL PERFORMANCE OF THE EVALUATED MODELS (TRIAL STRUCTURE IDENTIFICATION)

Model	Total	Correct	Partial	Wrong	Acc./Partial (%)
Llama3.1-8b	60	15	0	45	25.0/25.0
Deepseek-r1:8b	60	58	0	2	96.7/96.7
Qwen3-8b	60	36	18	6	60.0/75.0

Further analysis showed that the main limitation of Llama3.1-8b was the incorrect interpretation of descriptive clinical groups as formal cohorts. Phrases describing patient characteristics, such as disease stage or molecular status, were frequently interpreted as independent cohort structures, leading to false cohort identification.

For Qwen3-8b, most errors were related to schema compliance rather than semantic understanding. In several cases, the model correctly identified the absence of cohorts but returned incomplete JSON structures, omitting required fields such as the empty cohort array. Conversely, when explicit cohort structures were present, all models demonstrated substantially higher performance, indicating that the main challenge was distinguishing cohort absence from descriptive patient categories.

Beyond structural correctness, a qualitative expert assessment was conducted on a subset of four clinical trials (Trials 5, 16, 26, and 30) to evaluate whether generated cohort descriptions preserved clinically relevant information. The assessment focused on the maintenance of diagnosis, molecular alterations, previous treatment exposure, and therapeutic requirements. The summarized expert evaluation of the evaluated models is presented in Table V.

The expert evaluation demonstrated that correct cohort identification does not necessarily guarantee preservation of clinical meaning. Across the analyzed trials, the main errors were related to omission of eligibility-defining information rather than incorrect cohort structures.

TABLE V
EXPERT ASSESSMENT OF COHORT-STRUCTURE CLASSIFICATION PERFORMANCE

Model	Avg. Class.	Expert Assessment Summary
Deepseek-r1:8b	2.25	Correctly identifies cohort structures but frequently omits clinically relevant descriptors such as treatment status, molecular alterations, or diagnostic specificity.
Llama3.1-8b	2.75	Achieves high semantic fidelity in some trials but presents inconsistent behaviour, including complete loss of clinical information in specific cases.
Qwen3-8b	2.50	Correctly captures several cohort characteristics but occasionally omits essential molecular or therapeutic information.

Overall, these results indicate that cohort identification is primarily limited by the preservation of the complete clinical definition of each cohort. While structural extraction can be performed reliably, maintaining all clinically relevant eligibility characteristics remains a challenging aspect of LLM-based trial document understanding, as illustrated by the strict accuracy comparison between models (Figure 7).

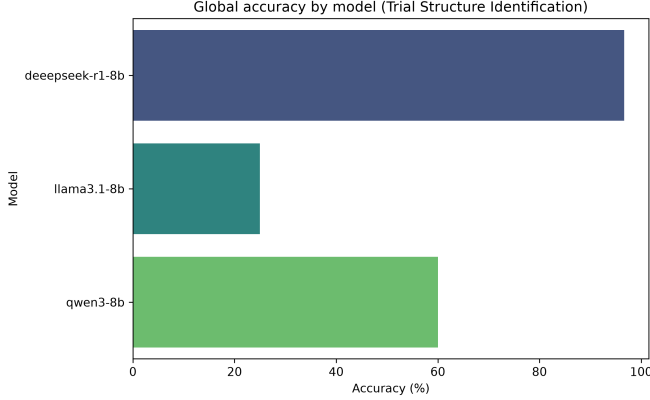


Fig. 7. Trial Structure Identification - Strict accuracy

Eligibility Criteria Extraction

The third experimental task evaluated the capability of the LLMs to extract individual clinical trial eligibility criteria while preserving their original clinical meaning and identifying whether each criterion was associated with a specific cohort. This step represents a fundamental component of the proposed pipeline, as accurate criteria extraction is required before their conversion into machine-interpretable rules and subsequent patient-trial matching.

The evaluation considered two complementary aspects: (i) the extraction of individual inclusion and exclusion criteria from clinical trial documents, and (ii) the attribution of extracted criteria to the corresponding cohort when cohort-specific eligibility requirements were present. Generated outputs were manually compared against manually annotated gold standards. Two metrics were reported: strict accuracy, considering only completely correct extractions, and partial accuracy, which also accounted for outputs that preserved the majority of the clinical meaning but contained minor structural or semantic deviations.

As presented in Table VI, substantial differences were observed between models. Qwen3-8b achieved the strongest overall performance, correctly extracting 823 out of 892 eligibility criteria, resulting in a strict accuracy of 92.3% and a partial accuracy of 95.9%. Furthermore, the model produced only four incorrect extractions, demonstrating a strong capability of preserving the original eligibility information while avoiding hallucinated criteria. Deepseek-r1:8b achieved intermediate performance, with strict and partial accuracies of 83.7% and 88.2%, respectively. In contrast, Llama3.1-

8b presented the lowest performance, achieving 71.9% strict accuracy and 75.8% partial accuracy, while also generating the highest number of incorrect criteria.

The second component of this task focused on identifying whether extracted criteria belonged to a specific trial cohort, with the corresponding results presented in Table VII. In this scenario, Qwen3-8b again demonstrated superior performance, correctly assigning cohort-specific criteria in 26 out of 27 cases, achieving 96.3% accuracy. Both Deepseek-r1:8b and Llama3.1-8b achieved 66.7% accuracy. The errors observed in these models were mainly associated with incomplete extraction rather than incorrect cohort attribution, suggesting that the main limitation was failure to preserve the complete criterion information before the classification step.

TABLE VI
GLOBAL PERFORMANCE OF THE EVALUATED MODELS (ELIGIBILITY CRITERIA EXTRACTION)

Model	Total	Correct	Partial	Wrong	Acc./Partial (%)
Llama3.1-8b	892	641	70	181	71.9/75.8
Deepseek-r1:8b	892	747	80	65	83.7/88.2
Qwen3-8b	892	823	65	4	92.3/95.9

TABLE VII
PERFORMANCE ON CRITERIA REQUIRING COHORT-STRUCTURE EXTRACTION

Model	Total	Correct	Partial	Wrong	Acc./Partial (%)
Llama3.1-8b	27	18	0	9	66.7/66.7
Deepseek-r1:8b	27	18	0	9	66.7/66.7
Qwen3-8b	27	26	0	1	96.3/96.3

The distribution of extraction outcomes and cohort assignment accuracy across models are further illustrated in Figures 8 and 9, respectively.

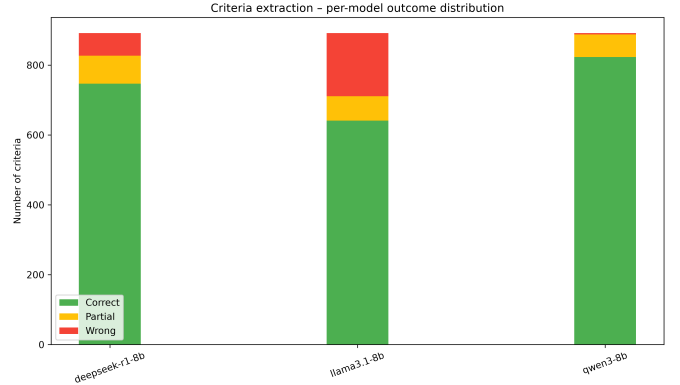


Fig. 8. Eligibility Criteria Extraction - Outcome distribution

Although the automatic evaluation provides an objective measurement of extraction performance across the complete dataset, clinical interpretation of eligibility criteria requires assessing whether deviations from the reference representation result in meaningful changes in clinical interpretation. Therefore, a subset of extracted criteria was additionally reviewed

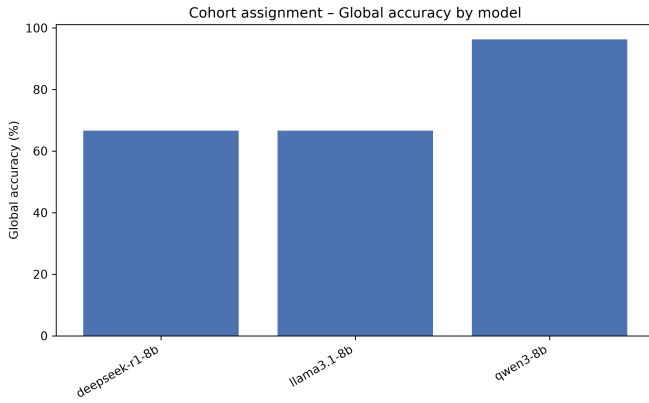


Fig. 9. Eligibility Criteria Extraction - Cohort assignment strict accuracy

by a clinical expert to characterize the severity and nature of possible errors.

The expert assessment confirmed the strong extraction capabilities observed in the automatic evaluation, particularly for Qwen3-8b and Deepseek-r1:8b. Across the evaluated trials, Qwen3-8b consistently preserved the clinical meaning of the eligibility criteria, with observed differences mainly related to acceptable restructuring or expansion of compound criteria. For example, when a broad criterion such as adequate organ function was divided into multiple individual laboratory constraints, the original clinical meaning was maintained and no clinically relevant information was lost.

Deepseek-r1:8b also demonstrated high clinical fidelity in the expert-reviewed subset, achieving complete preservation of inclusion and exclusion criteria in the evaluated trials. The expert analysis confirmed that the model was able to maintain compound requirements, avoid hallucinations, and preserve the relationship between multiple clinical constraints. These findings complement the quantitative results, where Deepseek showed intermediate but consistent extraction performance.

Llama3.1-8b showed a more variable behavior. Although the model successfully extracted most criteria, the expert analysis identified occasional structural errors that could affect downstream processing. In particular, one observed error consisted of splitting a compound tumor-size eligibility criterion into separate entries. Although all relevant information was preserved, this modification altered the original logical structure of the requirement and could potentially affect later conversion into machine-readable rules.

Overall, the expert validation indicated that most model deviations were related to representation differences rather than incorrect clinical interpretation. The main clinically relevant failure modes involved omission of eligibility components, incorrect fragmentation of compound criteria, or loss of relationships between conditions. These observations highlight the importance of maintaining a clinical validation step when transforming natural language eligibility requirements into

computational representations.

Eligibility Criteria Conversion

The fourth experimental task evaluated the capability of the LLMs to transform individually extracted clinical trial eligibility criteria into machine-interpretable logical representations while preserving their original clinical meaning. This step represents a critical component of the proposed pipeline, as the generated logical rules enable the deterministic matching engine to perform patient-trial eligibility assessment with reduced dependency on LLM-based reasoning during the final decision process.

The evaluation focused on the conversion of individual inclusion and exclusion criteria previously extracted from clinical trial documents. Generated logical representations were manually evaluated to identify if they kept the clinical meaning of the criteria. Two metrics were reported: strict accuracy, considering only fully correct conversions, and partial accuracy, accounting for cases where the majority of the clinical meaning was preserved but minor structural deviations or incomplete logical representations were present.

It is important to note that the number of evaluated converted criteria differed between models. This difference resulted from occasional generation failures, where models produced explanations, additional text, or invalid outputs instead of the required structured representation, despite explicit instructions to return only the converted criterion. These cases were considered conversion failures and excluded from the correctness assessment, while still reflecting the model’s ability to follow the required output format.

As shown in Table VIII, the conversion task represented a considerably more challenging problem compared with previous extraction tasks. Qwen3-8b achieved the highest strict accuracy, correctly converting 651 criteria, corresponding to 64.1% strict accuracy and 74.4% partial accuracy. However, it also produced the highest number of incorrect conversions (156), indicating that while the model was frequently capable of producing complete logical representations, failures tended to result in clinically incorrect rules rather than partially incomplete ones.

Llama3.1-8b achieved comparable performance, with 62.5% strict accuracy and 73.5% partial accuracy. Despite converting fewer criteria than Qwen3-8b, the model demonstrated a similar balance between correct and partially correct outputs. Deepseek-r1:8b presented the lowest strict accuracy (52.2%), although its partial accuracy increased substantially to 71.2%, indicating that many of its errors were related to incomplete logical formalization rather than complete misunderstanding of the eligibility requirement.

The distribution of correct, partial, and incorrect conversions across models is further illustrated in Figure 10.

A common failure pattern observed across all evaluated models was the difficulty in converting complex compound

TABLE VIII
GLOBAL PERFORMANCE OF THE EVALUATED MODELS (ELIGIBILITY
CRITERIA CONVERSION)

Model	Total	Correct	Partial	Wrong	Acc./Partial (%)
Llama3.1-8b	969	606	212	151	62.5/73.5
Deepseek-r1:8b	1172	612	444	116	52.2/71.2
Qwen3-8b	1016	651	209	156	64.1/74.4

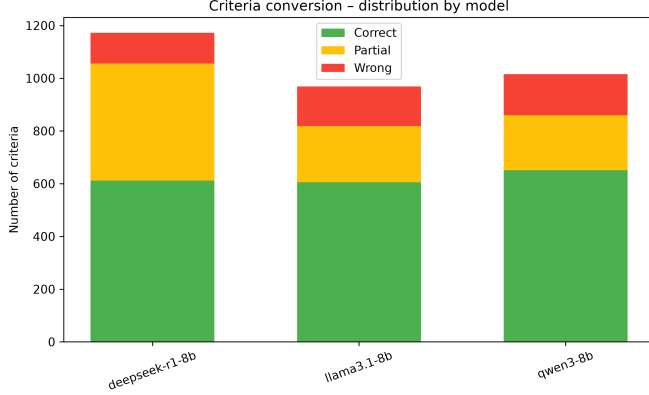


Fig. 10. Eligibility Criteria Conversion

criteria. Eligibility requirements containing nested logical relationships, particularly combinations of multiple AND/OR conditions, frequently resulted in partially represented rules. Instead of preserving the complete logical structure, models often extracted only the most prominent condition while omitting secondary constraints or incorrectly simplifying relationships between conditions.

Another common problem was related to the conversion of said fields to schema fields, specially to the treatment field, the models consistently failed to identify that certain criteria could easily be represented with the patient schema field of treatments, resulting in new, not schema included fields, that would force an LLM evaluation that wouldn't be necessary.

This limitation was particularly relevant because compound eligibility criteria are common in oncology clinical trials, where patient selection often depends on the simultaneous evaluation of diagnosis, molecular profile, previous treatments, laboratory values, and disease status. Therefore, although the models demonstrated reasonable capability in converting simpler eligibility constraints, additional improvements are required for reliably formalizing complex clinical decision rules.

Patient-Trial Matching

The final experimental task evaluated the capability of the LLMs to assess patient eligibility when confronted with clinical requirements that could not be directly represented within the predefined patient profile schema. This step represents the final decision-support component of the proposed pipeline, where the objective is to extend the deterministic

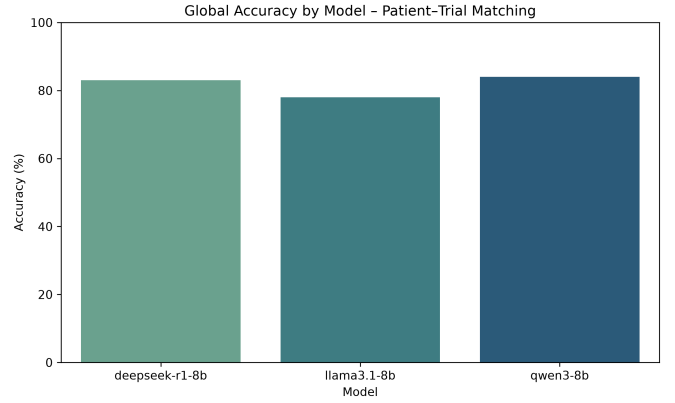


Fig. 11. Patient-Trial Matching - Strict accuracy by model

matching engine with LLM-based clinical interpretation while maintaining transparency and expert control.

In this scenario, the LLM receives the patient structured profile, the original clinical diary, relevant laboratory information, and an eligibility criterion represented as a logical rule. The model is then required to determine whether the available patient information satisfies the given criterion and provide a clinically grounded justification for the decision.

The evaluation considered two complementary aspects: (i) correctness of the eligibility assessment, manually evaluated, and (ii) quality of the generated justification. Since this task inherently involves clinical reasoning and interpretation of incomplete medical information, a complementary expert assessment was performed on a representative subset of generated outputs. The expert evaluation focused on clinical reliability, evidence grounding, conservative handling of missing information, and potential impact of incorrect decisions.

As shown in Table IX, all models demonstrated strong performance in the eligibility assessment task. Qwen3-8b achieved the highest quantitative performance, correctly evaluating 84 out of 100 criteria, corresponding to a strict and partial accuracy of 84.0%. Deepseek-r1:8b achieved a very similar result with 83.0% accuracy, while Llama3.1-8b obtained 78.0%. No partially correct classifications were observed in this task, as incorrect eligibility decisions generally resulted from an incorrect interpretation of the available clinical evidence rather than from partially satisfying the criterion.

TABLE IX
GLOBAL PERFORMANCE OF THE EVALUATED MODELS (PATIENT-TRIAL
MATCHING)

Model	Total	Correct	Partial	Wrong	Acc./Partial (%)
Llama3.1-8b	100	78	0	22	78.0/78.0
Deepseek-r1:8b	100	83	0	17	83.0/83.0
Qwen3-8b	100	84	0	16	84.0/84.0

The strict accuracy comparison between models is further illustrated in Figure 11.

Beyond quantitative accuracy, the expert assessment pro-

vided additional insight into the clinical reliability of the generated decisions. Overall, Deepseek-r1:8b and Qwen3-8b demonstrated highly consistent behaviour, with both models grounding their decisions in explicit information available in the patient documentation. These models correctly identified measurable lesions, laboratory thresholds, and molecular alterations when present, while appropriately considering missing information as insufficient evidence rather than assuming eligibility or ineligibility.

Qwen3-8b achieved the highest expert-rated performance, with an average justification score of 4.93/5. The model consistently demonstrated conservative reasoning, avoiding unsupported assumptions and correctly handling incomplete clinical information. In particular, it correctly interpreted measurable disease according to lesion size, laboratory thresholds such as ANC requirements, and molecular eligibility criteria. Missing values, such as unavailable creatinine clearance calculations, were appropriately identified as unknown rather than incorrectly classified.

Deepseek-r1:8b achieved an average expert score of 4.8/5, showing similar strengths. The model consistently produced evidence-based decisions and avoided hallucinating clinical information. Its main advantage was the ability to explicitly distinguish between confirmed absence of a criterion and unavailable information, which is particularly important in a clinical decision-support scenario.

Llama3.1-8b achieved an average expert score of 4.2/5. Although the model showed strong performance when information was explicitly documented, the expert evaluation identified occasional over-interpretation of uncertainty. Examples included incorrectly rejecting criteria despite available laboratory values satisfying the requirement or inferring unsupported conclusions from incomplete documentation. These errors were not caused by hallucinated information, but rather by excessive conservatism or speculative interpretation.

The expert-assigned scores for the generated eligibility justifications are summarized in Table X.

TABLE X
EXPERT-ASSIGNED AVERAGE SCORES FOR ELIGIBILITY JUSTIFICATION

Model	Expert Avg. Score
Deepseek-r1:8b	4.8/5
Llama3.1-8b	4.2/5
Qwen3-8b	4.93/5

Overall, these results suggest that small LLMs are capable of performing clinically aligned eligibility assessments when constrained by structured patient information and explicit evidence grounding. Importantly, the expert evaluation indicates that model reliability is not solely determined by classification accuracy, but also by the ability to justify decisions transparently and avoid unsupported clinical assumptions. This reinforces the proposed clinician-in-the-loop approach, where LLMs act as interpretable decision-support tools rather than autonomous clinical decision makers.

DISCUSSION

This pipeline demonstrated that small LLMs (8B parameters) are capable of performing relevant tasks in a complex clinical context, although with different levels of difficulty.

The results obtained present a clear tendency: models performed well in tasks of direct structured extraction, clearly faced their greatest challenges in the formalization of logic, where the worst results were observed across all models, and presented good results in tasks involving conditioned clinical reasoning when explicit context was available.

The performance of the selected trio of models was not homogeneous. Qwen3 clearly shined on extraction tasks, achieving the highest strict accuracy across both hyper-relevant parameter extraction (78.3%) and eligibility criteria extraction (92.3%). Deepseek distinguished itself in trial structure identification (96.7%) and presented strong conservative reasoning in patient-trial matching, while Llama presented the highest variability of results across all tasks.

These experiments showed that small models can support parts of a complex clinical pipeline when the task is properly decomposed and when there is expert oversight. The conclusion is not that one model is universally better, but rather that different models excel in different components, and a properly structured pipeline can leverage these complementary strengths.

Research Hypotheses Assessment

The results obtained allow a direct assessment of the four research hypotheses guiding this study.

H1 is supported by the experimental results. All three models demonstrated the ability to extract and normalize clinically relevant information from heterogeneous oncology diaries while achieving 100% schema compliance. Qwen3-8B achieved the strongest performance with 78.3% strict accuracy, confirming that structured extraction from free-text oncology documentation is feasible with small LLMs, though performance remains dependent on field complexity.

H2 is partially supported. Eligibility criteria extraction yielded strong results, with Qwen3-8B achieving 92.3% strict accuracy, and cohort attribution reaching 96.3% for the same model. However, the subsequent formalization of extracted criteria into machine-interpretable logical rules proved considerably more challenging, with strict accuracies ranging from 52.2% to 64.1% across models. H2 is therefore supported with respect to extraction and preservation of clinical meaning, but only partially supported with respect to the formalization of logical relationships.

H3 is supported by the patient-trial matching results. The hybrid mechanism demonstrated strong performance, with Qwen3-8B and Deepseek-R1-8B achieving 84.0% and 83.0% accuracy respectively, and expert-rated justification quality scores above 4.8 out of 5. The expert evaluation confirmed

that decisions were consistently grounded in available clinical evidence, supporting both the effectiveness and the interpretability of the proposed matching strategy.

H4 is broadly supported. Across all evaluated tasks, small open-source models (8B parameters) demonstrated sufficient performance to support the proposed workflow, particularly in extraction and matching tasks. The main exception remains logical formalization, where performance was more limited, suggesting that this specific component may benefit most from larger models or targeted fine-tuning in future work.

Strengths of the Proposed Approach

The results suggest that the proposed approach has significant potential as a clinical decision-support tool for patient recruitment in oncology clinical trials. One of its main strengths lies in its modular design. Rather than treating eligibility assessment as a single end-to-end task, the workflow decomposes the problem into a sequence of specialized stages, namely patient information extraction, trial structure identification, eligibility criteria extraction, criteria formalization, and patient-trial matching.

This decomposition reduces the complexity of individual tasks and allows LLMs to focus on well-defined objectives. Consequently, each component can be evaluated, validated, and improved independently, increasing the overall robustness of the system. As can be seen in the experimental results, it allowed even small models to obtain good results, where an end-to-end model would most likely have had serious difficulties, while also making it harder to identify where failures occurred in the first place. These individual evaluations allowed to understand that extraction is not equal to conversion, which also is not equal to matching. A model can be strong on one task and weak on another, Qwen's strength in extraction (92.3% strict accuracy on criteria extraction) and Deepseek's strength in structure identification (96.7%) are a clear example of this. This allows the pipeline to employ the best performing model on the specific task at which it excels, rather than relying on a single model globally.

Another important strength is the high degree of transparency and interpretability provided by the framework. Every intermediate artifact generated by the system, including structured patient profiles, extracted eligibility criteria, logical representations, and matching decisions, remains accessible for inspection. This enables clinicians to understand how eligibility decisions are reached rather than relying on opaque end-to-end model predictions. A system that provides correct outputs but does not explain how is not acceptable in a clinical setting. The expert validation confirmed that Qwen and Deepseek consistently grounded their decisions on available evidence, avoiding hallucinated data, with justification scores of 4.93/5 and 4.8/5, respectively. The clinical justification for a decision is as important as the classification accuracy itself, particularly when the objective is to reduce unnecessary screen time for clinicians.

The clinician-in-the-loop design further contributes to the reliability of the approach. Expert validation is incorporated throughout the workflow, allowing clinicians to review, correct, and approve intermediate outputs before they are used in subsequent stages. This not only increases clinical safety but also mitigates the impact of occasional LLM errors.

Finally, the semi-deterministic matching strategy reduces dependence on probabilistic reasoning by maximizing the use of structured data and rule-based evaluation whenever possible. Unlike many existing approaches that rely on LLMs to directly answer "Is the patient eligible?", the proposed system structures the patient profile, formalizes eligibility criteria into logical rules, and only invokes LLM-based reasoning when the deterministic engine cannot reach a conclusion. This reduces the risk of hallucination, increases explainability, and improves the clinical acceptability of the generated outputs. This approach improves consistency, reproducibility, and auditability while preserving the flexibility required to handle complex clinical eligibility requirements.

Challenges observed during LLM-based clinical processing

Structured extraction vs semantic interpretation

On simple and explicit fields, these models presented excellent results, gender and ECOG-PS achieved 100% accuracy across all three models, whereas on complex fields they faced considerable difficulties, particularly with normalized diagnosis, treatments, dates, and staging. Notably, diagnosis extraction was the most challenging field, with the best result being 46.7% achieved by Deepseek, and a recurrent error pattern in staging where models extracted only the final stage classification (e.g., "IVB") instead of the complete TNM representation (e.g., "cT3N2M1b (IVB)").

These challenges were not solely due to difficulties in extracting the text itself. They were also a consequence of the difficulty in following complex normalization mappings, adhering to clinical ontologies, and performing temporal interpretation, tasks that go well beyond surface-level text extraction.

Preservation of clinical meaning is harder than extraction

The results also showed that models would consistently identify cohort structures in clinical trials, but would lose clinically relevant details when extracting their descriptions. The expert assessment of trial structure confirmed this: despite Deepseek achieving 96.7% structural accuracy, it received an average clinical classification of only 2.25/5, frequently omitting treatment status, molecular alterations, or diagnostic specificity. Similarly, criteria extraction presented strong results across models, while their conversion into logical rules was considerably worse.

The main difficulty was not identifying information, but preserving the relationships between clinical concepts. Compound criteria such as "KRAS G12C inhibitor-naïve", "prior FGFR inhibitor exposure", or "primary platinum refractory disease"

involve nested clinical dependencies that models frequently simplified or partially omitted.

Logical formalization remains the bottleneck

The logical formalisation of extracted criteria was clearly the main challenge for these small models. While in some cases criteria extraction reached above 90% strict accuracy, Qwen achieving 92.3%, the conversion of these criteria into machine-interpretable logical rules dropped to approximately 64% for the same model. This gap is particularly significant because the conversion step is precisely what enables the deterministic matching engine to operate without LLM-based reasoning.

Small LLMs are effective at interpreting natural language, but face consistent challenges when transforming clinical language into formal logic. They consistently failed to completely represent AND/OR nested rules, dependencies between criteria, and compound eligibility requirements containing multiple simultaneous conditions. Another recurring failure was the inability to map certain criteria to existing schema fields, particularly the treatment field, resulting in the creation of new, schema-excluded fields that would then require additional LLM evaluation, undermining one of the main objectives of the pipeline. This limitation justifies and reinforces the hybrid architecture proposed in this work.

Limitations

Despite the promising results, the proposed approach presents several limitations.

First, as with most LLM-based systems, output quality is highly dependent on the quality and completeness of the input documentation. Missing, ambiguous, or poorly documented clinical information can negatively affect patient profile extraction and, consequently, all downstream tasks.

Second, the workflow follows a sequential architecture, meaning that errors introduced in earlier stages may propagate to subsequent components. Although the decomposed approach of the pipeline and expert validation substantially reduce this risk, they cannot eliminate it entirely, which still proves to be an advantage over an end-to-end approach. The overall performance of the matching process therefore remains partially dependent on the quality of preceding extraction and conversion steps.

Another limitation arises from the formalization of eligibility criteria. While many criteria can be represented through deterministic logical rules, highly complex clinical requirements often involve contextual reasoning, implicit assumptions, or nuanced clinical judgement. Such criteria may be difficult both for LLMs to formalize and for clinicians to validate within a structured representation.

Furthermore, the effectiveness of the matching process depends on the completeness of the extracted patient profile. When relevant information is absent from the clinical diary,

eligibility assessment may become inconclusive, limiting the usefulness of the generated recommendations and potentially increasing the amount of manual review required.

The expert validation, although valuable, was conducted on a reduced subset of the dataset and does not substitute a prospective clinical validation. This limits the strength of conclusions drawn from qualitative assessments and highlights the need for more extensive expert evaluation in future work.

Finally, the evaluation was conducted using synthetic clinical diaries and a limited number of clinical trials. Although considerable effort was made to maximize diversity and realism, performance on larger and fully real-world datasets may differ from the results observed in this study.

Clinical Implications

The proposed framework addresses a clinically relevant challenge: the identification of potentially eligible patients for oncology clinical trials. This process is traditionally time-consuming and requires clinicians to manually compare extensive free-text clinical documentation against often complex eligibility criteria.

By automatically extracting relevant patient information, structuring eligibility requirements, and providing preliminary matching assessments, the system has the potential to reduce cognitive burden and administrative workload. This may allow clinicians to dedicate more time to direct patient care and clinical decision-making.

Importantly, the framework was designed as a decision-support system rather than a decision-making system. Final responsibility remains entirely with the clinician, who can inspect, validate, and modify every intermediate result and matching decision. In this sense, the LLM serves as an assistive technology that summarizes and organizes information rather than replacing clinical expertise.

The traceability of all generated artifacts further supports clinical adoption by providing a complete audit trail from the original documents to the final eligibility recommendation.

The entire system was continuously evaluated by oncologists from the Oncology Department of Coimbra University Hospital Centre (HUC) through review meetings. These sessions allowed emerging issues and clinical needs to be promptly addressed, ultimately leading to a high level of clinician satisfaction with the system.

Generalizability

Although the study was conducted in collaboration with the Oncology Department of Coimbra University Hospital Centre (HUC), the proposed methodology was not designed specifically for a single institution.

The patient profile schema captures clinically meaningful concepts commonly used in oncology practice, while

the extraction and matching pipelines operate independently of institution-specific document layouts. Consequently, the framework could potentially be applied to clinical documentation originating from other hospitals, provided that the required information is present within the source documents.

The approach is also largely language-agnostic. Since the underlying methodology relies on prompt-guided information extraction and logical formalization, adaptation to other languages would primarily require modifications to the prompts and validation procedures rather than fundamental architectural changes.

Beyond oncology, the same methodological principles could potentially be extended to other medical specialties where patient recruitment for clinical studies depends on the evaluation of structured eligibility criteria against heterogeneous clinical documentation.

Future Work

Several opportunities exist for extending and improving the proposed framework.

First, this study focused exclusively on prompt engineering techniques. Future work could investigate parameter-efficient fine-tuning approaches, such as PEFT methods, as well as Retrieval-Augmented Generation (RAG) architectures to improve both extraction accuracy and eligibility assessment performance.

Second, although the prompts employed throughout the study underwent extensive manual refinement, they were not systematically optimized. More rigorous prompt optimization strategies could potentially yield further improvements across all evaluated tasks.

Third, the patient profile schema was defined collaboratively with the clinical stakeholder to capture the information considered most relevant for eligibility assessment. Future studies may explore richer and more comprehensive representations incorporating additional clinical variables, laboratory measurements, imaging findings, and longitudinal patient information.

Another important direction involves evaluating larger and more capable language models. Due to hardware constraints, the experiments were limited to open-source models with approximately eight billion parameters. Evaluating larger models may provide insights into the trade-off between computational requirements and performance gains in clinical eligibility assessment tasks.

Finally, future validation should be conducted on larger collections of real-world clinical documentation and clinical trials to further assess the robustness, scalability, and clinical applicability of the proposed approach in routine healthcare environments.

CONCLUSION

A. Contributions

This study presents a modular LLM-based pipeline for the identification of potentially eligible patients for clinical trials. The proposed approach decomposes the complex trial-patient matching problem into a sequence of specialized tasks, including the extraction and structuring of patient profiles from free-text clinical diaries, the identification and conversion of clinical trial eligibility criteria into machine-interpretable logical representations, and the final semi-deterministic patient-trial matching process.

The experimental evaluation demonstrated that small open-source LLMs (8B parameters) are capable of performing clinically relevant tasks across all pipeline components, though with varying levels of performance depending on task complexity. Structured extraction tasks yielded the strongest results, with Qwen3-8B achieving 78.3% strict accuracy in patient profile extraction and 92.3% in eligibility criteria extraction. Trial structure identification also produced strong results, particularly for Deepseek-R1-8B, which achieved 96.7% accuracy. In contrast, the logical formalization of eligibility criteria proved to be the most challenging task across all models, with strict accuracies ranging from 52.2% to 64.1%, highlighting that transforming clinical language into formal logic remains an open problem for small LLMs. Despite this, the final patient-trial matching stage demonstrated that when grounded in structured patient information and explicit clinical context, these models can perform reliable eligibility assessments, with Qwen3-8B and Deepseek-R1-8B achieving accuracy scores of 84.0% and 83.0% respectively, and expert-rated justification quality scores above 4.8 out of 5.

The modular design of the proposed methodology enables increased transparency, traceability, and interpretability of the generated outputs, which are critical requirements in clinical decision-support systems. By translating clinical trial criteria into logical representations and combining deterministic evaluation with targeted LLM assistance only when required, the system aims to reduce the dependence on the probabilistic reasoning capabilities of LLMs while preserving their flexibility in handling complex clinical information. Importantly, the evaluation confirmed that decomposing the problem into specialized stages not only improved performance but also allowed the identification of where specific failures occurred, something that would not be possible in an end-to-end approach.

Additionally, to enable a realistic evaluation under data privacy constraints, a clinically validated synthetic dataset was developed based on real oncology clinical documentation patterns, avoiding privacy risks while maintaining clinical realism. This dataset allowed the assessment of lightweight open-source language models under limited computational resources, reflecting realistic deployment constraints in healthcare environments.

B. Impact

The proposed system was designed with direct consideration of the needs and workflow constraints of clinical oncologists, positioning the clinician as the final decision-maker rather than replacing expert judgment. Through the incorporation of validation stages, persistent storage of intermediate outputs, and access to the reasoning behind LLM-assisted decisions, the pipeline provides a transparent and auditable framework suitable for clinical decision support. The expert validation conducted throughout the study, including qualitative assessment of clinical justifications, confirmed that the system's outputs are grounded in available evidence and avoid unsupported clinical assumptions.

By reducing the manual effort required to compare extensive clinical documentation against complex trial eligibility criteria, this approach has the potential to decrease clinician cognitive workload and facilitate more efficient identification of suitable clinical trials for patients.

C. Future Work

Although the proposed methodology demonstrates the feasibility of using lightweight LLMs within a clinical trial matching pipeline, several opportunities for further improvement remain. Due to the time and computational constraints of this study, the evaluation was limited to a reduced number of models and relatively small-scale language models. Future work should investigate the impact of larger and more capable models, as well as more advanced adaptation techniques such as parameter-efficient fine-tuning and retrieval-augmented generation. The observed gap between extraction and logical formalization performance also suggests that targeted fine-tuning on criterion conversion tasks could yield meaningful improvements.

Furthermore, evaluation with larger and more diverse clinical datasets, including real-world anonymized clinical documentation, would provide additional validation of the system's robustness and generalizability. Finally, integration with hospital information systems represents an important future direction, requiring secure infrastructures capable of supporting the processing of real clinical data while maintaining privacy and regulatory compliance.

Overall, this study demonstrates the feasibility of combining structured representations, deterministic logic, and lightweight LLMs to support transparent and clinically aligned patient-trial matching workflows, while also identifying logical formalization as the key remaining bottleneck for future development.

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